

11/20

condition

Q 1. Write a program to check whether a number is positive, negative, or zero

```
num = float(input("Enter a number:"))
if num > 0:
    print("The number is positive")
elif num < 0:
    print("The number is negative")
else:
    print("The number is zero")
```



Enter a number: 25

The number is positive

Q 2. Write a program to find the largest of three numbers.

```
a = int(input("Enter first number:"))
b = int(input("Enter second number:"))
c = int(input("Enter third number:"))
if a >= b and a >= c:
    largest = a
elif b >= a and b >= c:
    largest = b
else:
    largest = c
print("largest number is:", largest)
```



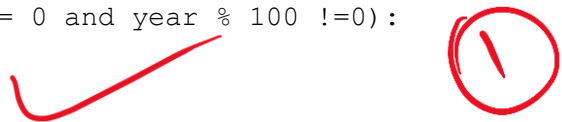
Enter first number: 80
Enter second number: 40
Enter third number: 50

largest number is: 80

Q3. Write a program to check whether a given year is a leap year.

```
year = int(input("Enter a year:"))

if (year % 400 == 0) or (year % 4 == 0 and year % 100 != 0):
    print("leap year")
else:
    print("Not a leap year")
```



Enter a year: 1990

Not a leap year

Q 4. Write a program to check if a character is a vowel or consonant.

```
str = input("Enter a character:").lower()

if str in "aeiou":
```



```
print("The character is avowel")
elif str.isalpha():
    print("The character is a consonant")
else:
    print(" please enter a valid character")
```

Enter a character: h

The character is a consonant

Q 5. Write a program to calculate the electricity bill:

- First 100 units → ₹5/unit
- Next 100 units → ₹7/unit
- Above 200 units → ₹10/unit

```
units = int(input("Enter units :"))
if units <= 100:
    bill = units * 5
elif units <= 200:
    bill = (units * 7) - 200
else:
    bill = (units * 10) - 800
print("Total bill:", bill)
```

Enter units : 400

Total bill: 3200

Loops (for / while)

1. Print all even numbers from 1 to 100.

```
for i in range(1, 101):
    if i % 2 == 0:
        print(i)
```

2
4
6
8
10
12
14
16
18
20
22
24
26
28
30
32
34
36
38
40
42

44
46
48
50
52
54
56
58
60
62
64
66
68
70
72
74
76
78
80
82
84
86
88
90
92
94
96
98
100

2. Find the factorial of a given number using a loop.

```
num = int(input("Enter a number:"))  
fact = 1
```

```
for i in range(1, n + 1):  
    fact = fact * i  
    print("factorial of", n, "is:", fact)
```

Enter a number: 8

```
factorial of 5 is: 1  
factorial of 5 is: 2  
factorial of 5 is: 6  
factorial of 5 is: 24  
factorial of 5 is: 120
```

3. Print the Fibonacci series up to n terms.

```
n = int(input("Enter a number:"))  
a = 0  
b = 1  
i = 0  
print("fibonacci series:")  
while i < n:  
    print(a, end=" ")  
    c = a + b  
    a = b
```

$a, b = b, a + b$

```
b = c
i = i + 1
```

Enter a number: 6

fibonacci series:
0 1 1 2 3 5

4. Find the sum of digits of a given number.

```
n = input("Enter a number:")
sum = 0
for digit in n:
    sum = sum + int(digit)
print("sum of digits:", sum)
```

Enter a number: 987

sum of digits: 9
sum of digits: 17
sum of digits: 24

5. Check whether a number is an Armstrong number

Functions

1. Write a function to find the greatest of three numbers.

```
def greatest(a, b, c):
    return max(a, b, c)
num1 = int(input("Enter first number:"))
num2 = int(input("Enter second number:"))
num3 = int(input("Enter third number:"))
print("Greatest number is:", greatest(num1, num2, num3))
```

Enter first number: 11
Enter second number: 45
Enter third number: 50

Greatest number is: 50

2. Write a function to check whether a number is prime.

```
def is_prime(n):
    if n <= 1:
        return False
    if n == 2:
        return True
    if n % 2 == 0:
        return False
    for i in range(3, n):
        if n % i == 0:
            return False
```

3. Write a function that returns the reverse of a string.

```
def is_palindrome(s):
```

```

reverse = s[::-1]
if s == reverse:
    return True
else:
    return False
word = input ("Enter a string:")
if is_palindrome(word):
    print(word, " is a palidrome")
else:
    print(word, "is not a palindrome")

```

Enter a string: madam

madam is a palidrome

4. Write a function to calculate the area and perimeter of a rectangle.

5. Write a function that accepts a list and returns the second largest element

excepcion handling

1. Write a program to handle ZeroDivisionError.

2. Write a program to handle ValueError when the user enters invalid

```

try:
    num = int(input("Enter a number:"))
    print("you entered:", num)

```

```

except ValueError:
    print("Invalid input")

```

Enter a number: 5

Invalid input

3. Create a custom exception to check if age is less than 18.

4. Write a program that handles both IndexError and KeyError.

5. Write a program using try, except, else, and finally to read a file and display its contents.